

**Team Members:**

| No. | Full name          | Role                            |
|-----|--------------------|---------------------------------|
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**1. PROJECT OVERVIEW**

**1.1 Project Title:** Moss-based Air Purification System

**1.2 Key challenge areas:**

☒ Track 2: Urban Air Quality and Climate Resilience

**1.3 Solution Category:**

☒ Integrated Solution (Hardware + Software)

**1.4 Project Summary:**

Our project proposes an intelligent biological air purification system called the “Moss-based Air Purification System.” It addresses the severe health threats posed by fine particulate matter (**PM2.5 and PM10**) in urban areas. By integrating living moss with IoT and AI technologies, the device provides a sustainable, eco-friendly alternative to conventional mechanical purifiers. Key technologies include real-time environmental sensors, automated microclimate control, and Edge AI for moss health monitoring. The system is designed to benefit urban residents and Smart Home users by promoting a green living experience and reducing dependence on non-biodegradable filtration materials.

**2. CONTEXT AND PROBLEM STATEMENT**

Air pollution, especially fine particulate matter (PM2.5/PM10) carrying toxic heavy metals from traffic and industry, causes over 8 million annual deaths and severe respiratory and cardiovascular diseases. This crisis is critical in rapidly developing cities where indoor and outdoor air quality frequently hits hazardous levels. The most affected stakeholders are urban residents, office workers, and vulnerable groups like children and the elderly who spend most of their time indoors.

**3. EXISTING SOLUTIONS AND GAP ANALYSIS**

Current solutions such as HEPA air purifiers and urban greenery face significant limitations:

- **HEPA Purifiers:** Highly effective but rely entirely on non-biodegradable synthetic materials, require frequent filter replacements, and consume significant energy.
- **Existing Moss Walls:** Moss has been recognized for its superior particulate absorption capability due to its unique biological structure. However, existing moss wall systems often rely on synthetic materials that may generate microplastics, while moss itself struggles to survive stably under harsh indoor climatic conditions.

**Gap Analysis:** There is a growing need to develop an intelligent biological air purification system utilizing moss integrated with microclimate control technology. This fills the gap

between static green walls (which lack environmental control) and mechanical purifiers (which lack biological sustainability).

## 4. PROPOSED SOLUTION AND CORE FEATURES

The project proposes the development of an intelligent biological air purification system. Unlike conventional air purifiers that rely entirely on mechanical filtration systems, this device utilizes living moss (*Hypnum cupressiforme*, *Sphagnum moss*, *Dicranum scoparium*) as the primary biological filtration component. Moss absorbs fine particulate matter, captures CO<sub>2</sub> through photosynthesis, and naturally maintains indoor humidity.

### Core Features:

- 1. Biological Filtration:** Utilizes moss and beneficial microorganisms (*Bacillus subtilis*, *Lactobacillus spp.*) to absorb PM2.5/PM10, decompose organic odors, and inhibit mold growth.
- 2. Automated Microclimate Control:** Integrates sensors to control ultrasonic mist generators, mini water pumps, and LED grow lights, ensuring optimal living conditions for the moss.
- 3. Real-time AIoT Monitoring:** Uses an ESP32-S3 controller to synchronize data to Firebase, allowing users to monitor air quality (PM2.5, PM10, temperature, humidity) via a mobile application.
- 4. Adaptive Control:** Automatically adjusts fan speed, LED intensity, and misting levels based on real-time sensor readings to optimize both air purification and moss health.

## 5. INNOVATION & COMPETITIVE ADVANTAGE

### Innovation & Competitive Advantage:

The project's innovation lies in the use of living moss as a biological filtration material integrated with AIoT technologies and sustainable design principles. Rather than functioning solely as a conventional air purifier, the system is designed as an "Environmental Wellness Device" that creates a green, relaxing, and user-friendly living environment. This concept aligns with the growing trends of Biophilic Design and modern Smart Home ecosystems.

### Feasibility & Development Plan:

- **Hackathon Phase:** Integrate the ESP32-S3 with PMS7003 and SHT31 sensors, establish the control logic for the ventilation and misting mechanisms, and successfully transmit real-time data to a Firebase cloud environment.
- **Post-Hackathon Roadmap:** Implement a lightweight YOLO or TinyML model deployed on an Edge AI device to monitor moss coloration and predict dehydration or decay. Optimize the aerodynamic design of the filtration chamber for commercial production.

## 6. TARGET GROUPS & POTENTIAL IMPACT

### Target Audience:

- Urban households seeking sustainable Smart Home devices.
- Offices, cafés, hospitals, and schools aiming to improve indoor air quality and aesthetics.

### Potential Impact:

The project aims to improve indoor air quality through an intelligent biological filtration solution. Studies have shown that moss-based filtration membranes can achieve an average

filtration efficiency of approximately 41% for PM2.5 and 47% for PM10 (up to 56% under specific conditions). Furthermore, the system increases natural humidity levels, reduces dryness caused by air conditioning, and creates a relaxing green environment. If scaled, this model could be expanded into a network of biological filtration devices for smart cities.

## 7. DESCRIPTION OF TECHNOLOGIES APPLIED

### A. Proposed Technologies

- **Hardware:** ESP32-S3, PMS7003 (Dust), SHT31 (Temp/Humidity), GY-30 (Light), OLED Display, LM2596 buck converter, 12V adapter, PWM fans, Ultrasonic mist generators, UV lighting.
- **Software & Programming:** C/C++ (Hardware control), Python (Data processing/AI training), Flutter (Mobile App), Firebase (Cloud Database).
- **AI/Machine Learning:** YOLO architecture for Edge AI image processing (Moss health monitoring).

### B. System Architecture & Technical Approach

The system operates on an Edge-to-Cloud architecture.

1. **Data Collection:** PMS7003, SHT31, and GY-30 sensors continuously gather environmental data.
2. **Edge Processing:** The ESP32-S3, programmed in C/C++, acts as the central hub. It uses Adaptive Control algorithms to locally trigger actuators (fans, water pumps, misting systems, LEDs) based on sensor thresholds.
3. **Vision & AI Integration:** A camera module feeds images to a lightweight YOLO model to classify the moss status (healthy, dry, decaying).
4. **Cloud & User Interface:** Data is pushed in real-time to Firebase. The Flutter mobile app fetches this data, providing users with a dashboard, push notifications for low water levels, and maintenance alerts.

### C. Data and Infrastructure Requirements

- **Data Sources:** Real-time sensor telemetry and scheduled image captures of the biological filter.
- **Infrastructure:** Wi-Fi network for ESP32 connectivity, Firebase for NoSQL database and authentication, and a local edge-processing unit for running the TinyML/YOLO inference without heavy cloud computing costs.

## 8. REFERENCES

- [1] A. Biloshchytskyi et al., "Reducing Outdoor Air Pollutants through a Moss-Based Biotechnological Purification Filter in Kazakhstan," *Urban Sci.*, vol. 7, no. 4, p. 104, 2023. Available: <https://doi.org/10.3390/urbansci7040104>
- [2] R. Alvarez Gutiérrez, J. Blom, B. Belmans, A. De Bock, L. Van den Bergh, and A. Audenaert, "Preliminary Research on Moss-Based Biocomposites as an Alternative Substrate in Moss Walls," *Sustainability*, vol. 15, no. 23, p. 16500, 2023. Available: <https://doi.org/10.3390/su152316500>
- [3] P. Lazo, A. Kika, F. Qarri, L. Bekteshi, S. Allajbeu, and T. Stafilov, "Air Quality Assessment by Moss Biomonitoring and Trace Metals Atmospheric Deposition," *Aerosol and Air Quality Research*. Available: <https://doi.org/10.4209/aaqr.220008>